生醫量測系統設計
(9920ET5610701)
Design of Biomedical Measurement Systems

Blood pressure and sound

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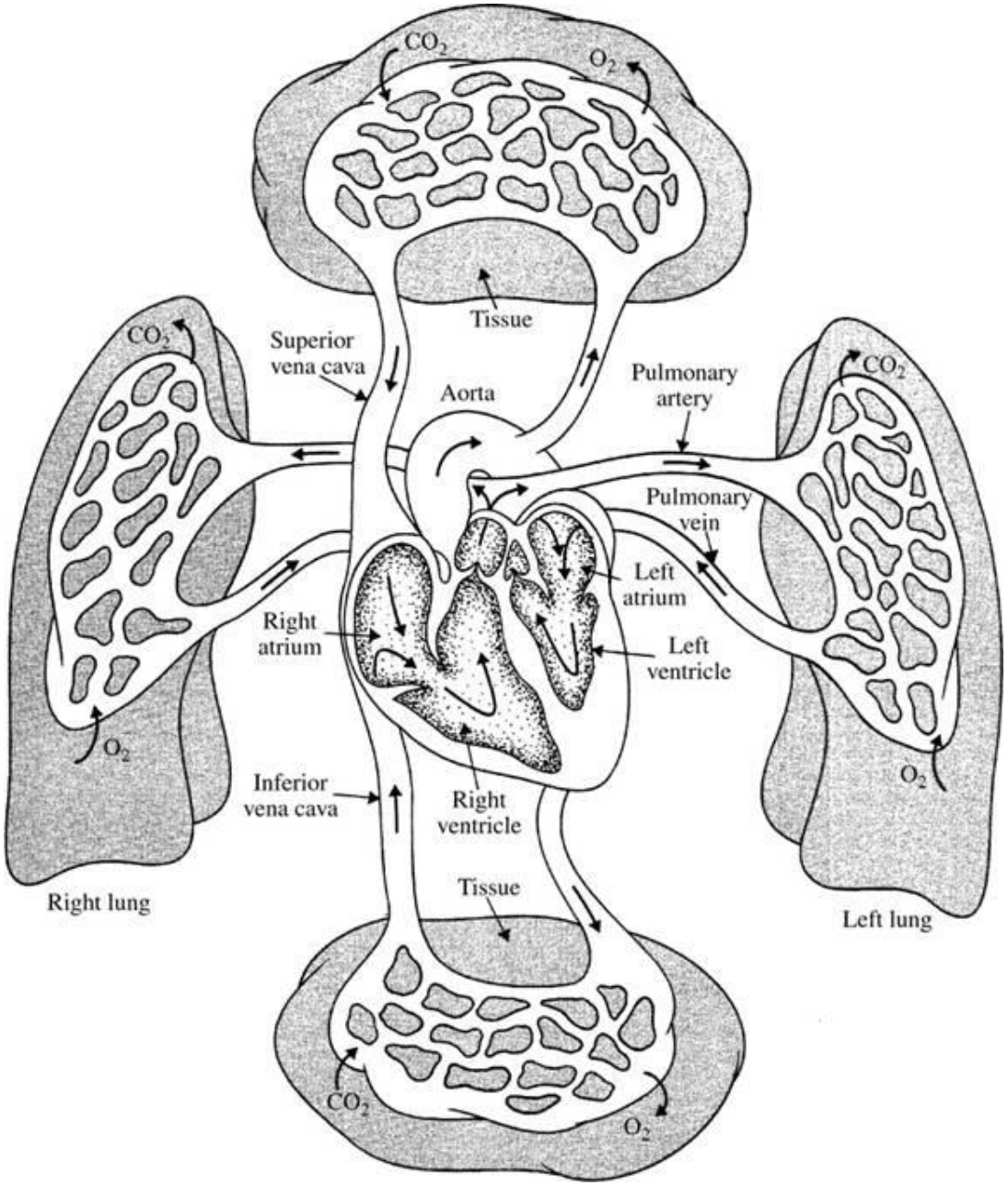
The content of this ppt is referred to Medical Instrumentation Application and Design (Hardcover) by John G. Webster, Fourth Edition

Blood pressure and sound

Figure 7.1 The left ventricle ejects blood into the systemic circulatory system. The right ventricle ejects blood into the pulmonary circulatory system.



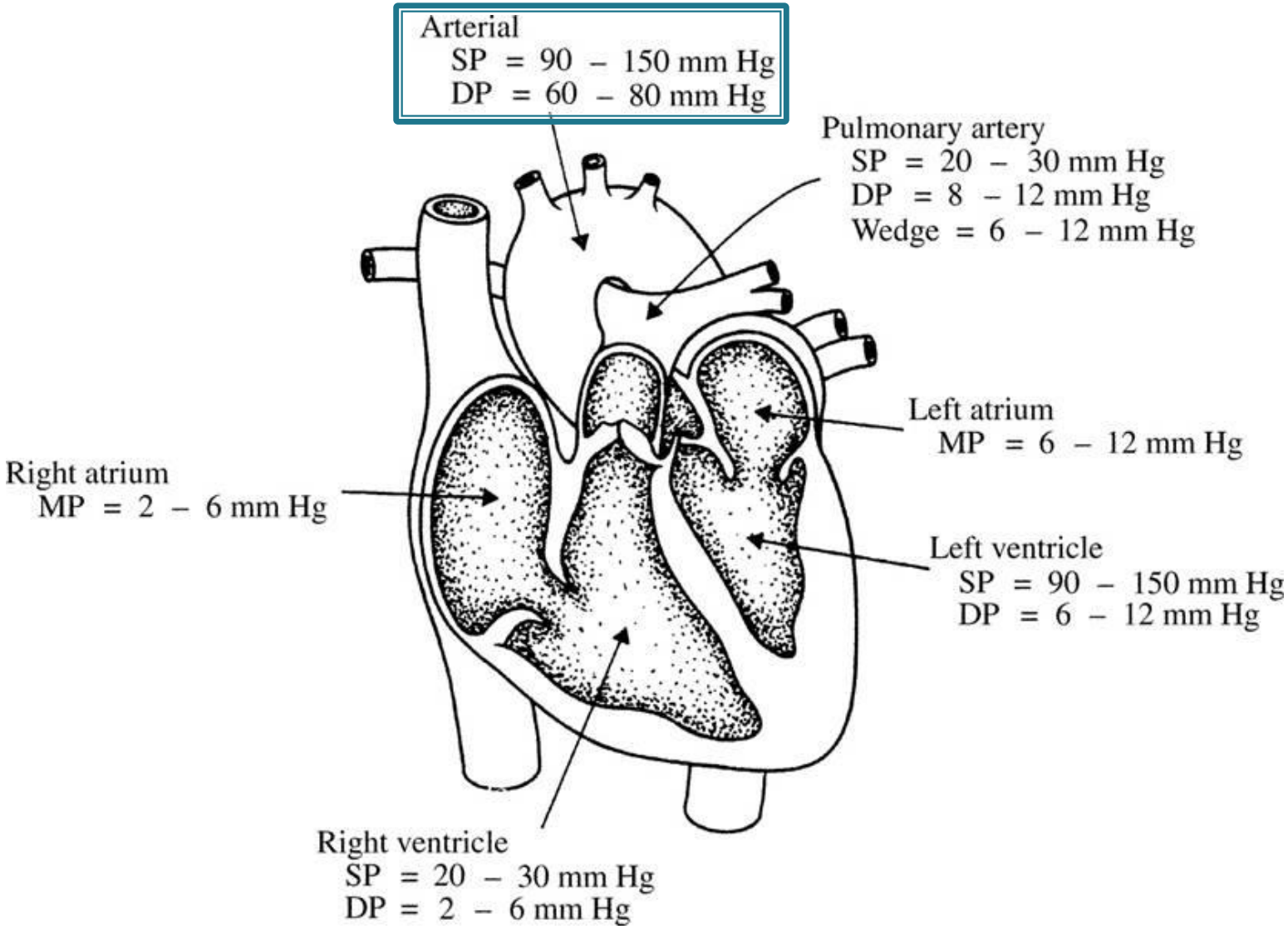


Figure 7.2 Typical values of circulatory pressures SP is the systolic pressure, DP the diastolic pressure, and MP the mean pressure.

動脈血壓的定義

- 是指當心臟收縮和舒張時，血液對動脈管壁造成的波動性壓力。
- 血壓是隨著心臟的博動而變動，心臟收縮時血管內有最大的血液量，此時的血壓稱收縮壓。心臟舒張時血管內有最小的血液量，此時的血壓稱舒張壓。收縮壓與舒張壓之差距稱為脈搏壓。
- 平均動脈壓 = 1/3脈搏壓 + 舒張壓
  = 1/3收縮壓 + 2/3舒張壓

Refer to
http://hospital.kingnet.com.tw/service.php?mob=modload&name=diyonline&file=3-11

影響血壓高低之因素

- 1.心搏出量
- 2.末梢血管之阻力
- 3.血管壁之彈性
- 4.血管之反應性
- 5.血液循環量
- 6.血液之黏稠度

Refer to
http://hospital.kingnet.com.tw/service.php?mob=modload&name=diyonline&file=3-11

血壓之調節

- 主要來自神經系統功能與體液中激素。
- 血壓調節的神經中樞位於延髓的血管運動中樞及心臟中樞。
- 動脈中含氧及二氧化碳的濃度、血壓值的高低、情緒和痛覺刺激及胸腔內壓等將刺激血管運動中樞及心臟中樞，影響血壓的上升或下降。

Refer to
http://hospital.kingnet.com.tw/service.php?mob=modload&name=diyonline&file=3-11

其他生理因素影響血壓

- 1.年齡：年齡越大，動脈彈性越差，血壓有上升之傾向
- 2.性別：一般男性的血壓比女性的血壓高出5-10mmHg
- 3.溫度：氣溫較高處血管擴張，血壓較降；氣溫較低處血管收縮，血壓較　高。體溫高時血管擴張，血壓也降；但惡寒顫抖時血管收縮，血壓上升。
- 4.新陳代謝：代謝率亢進時血壓上升，代謝率下降時血壓下降。
- 5.姿勢：姿勢對於血壓的影響來自於重力，平躺時收縮壓最高，其次為坐姿，其次為立姿。而舒張壓則反之，立姿時最高，其次為坐姿，其次為臥姿。
- 6.左右手差：由於右鎖骨下動脈比左鎖骨下動脈接近主動脈出心臟端，正常情形下，右手血壓比左手血壓高約5-10mmHg，但超出此範圍時則屬不正常。
- 7.上下肢差：採立姿量上下肢血壓時，通常下肢高10-15mmHg，超過20mmHg時，或下肢反比上肢低時，為不正常之現象。

Refer to
http://hospital.kingnet.com.tw/service.php?mob=modload&name=diyonline&file=3-1

血壓值代表意義

- 世界衛生組織定血壓的正常值為收縮壓140mmHg以下，舒張壓90mmHg以下，當收縮壓高於160mmHg，舒張壓高於95mmHg時，稱為高血壓。
- 而收縮壓在140-160mmHg間，舒張壓在90-95mmHg間為臨界值。
- 至於收縮壓低於90-100mmHg時，稱為低血壓。
- 臨床上，將舒張壓在90-104mmHg者稱為輕度高血壓，舒張壓在105-114mmHg者稱為中度高血壓，而舒張壓在115mmHg以上者，稱為重度高血壓。

Refer to
http://hospital.kingnet.com.tw/service.php?mob=modload&name=diyonline&file=3-11

- 衛生署修訂最新的血壓標準比過去更加嚴格，收縮壓與舒張壓在120到80之間才算正常，其中還把收縮壓120到130的列為高危險群，標準變嚴格後，衛生署估計，現在高血壓危險群可能會有5、600萬人！
- 高血壓防治需要更早一點！根據國民健康局新修訂的「高血壓防手冊」，增列了高血壓前期定義，只要收縮壓 120至 139mmHg、舒張壓 80至 89 mmHg的民眾，就算是高血壓危險群，雖然不用服藥，但要開始著手預防。
- 過去民眾總認為血壓是超過 140/90以上，才算是高血壓，這次修訂血壓標準最大變動，就是在正常血壓及第一、二期高血壓前，增列高血壓前期，此時血壓已算異常，需要靠健康促進如良好生活型態、飲食習慣以及戒菸、多運動來降低風險。
- 國民健康局成人及中老年保健組長徐瑞祥指出，過去高血壓主要分為第一、二期，第一期是指收縮壓 140至 159、舒張壓 90至 99，第二期為收縮壓 160以上、舒張壓 100以上，第一期病人需要服用藥物，第二期病人則要用2種以上降血壓藥才能控制血壓。
- 徐瑞祥說，這次在一、二期前增列高血壓前期的最大目的，就是希望醫師在遇到這類病患時，就要開始衛教病人，告訴這些高血壓的危險群，僅管還不用吃降血壓藥物，但要有危機意識，從生活中開始調整，才能預防高血壓。

Refer to 網路新聞

20歲開始自我健康管理

- 根據國民健康局民國91年「台灣地區高血壓、高血糖、高血脂盛行率調查」資料顯示，台灣地區40歲以上的成年人中，35%患有高血壓，65歲以上高血壓的比率更高達50%，而19歲以上的民眾，也有22.7%已經有高血壓，這讓防治工作，不再只著眼於老人，而是從年輕人開始，全面防治，目前宣導對象已經提前到20歲。國民健康局希望及早推「自我健康管理」的概念，讓民眾對自己的健康負責，養成適量喝酒、台灣地區40歲以上的成年人，35%有高血壓，比率相當高，建議應從20歲開始預防高血壓。

  從日常生活中，懂得自我健康管理：少喝酒，戒菸，規律運動，健康飲食，身體自然老當益壯。

  不抽菸、規律運動、健康飲食等...健康生活型態。

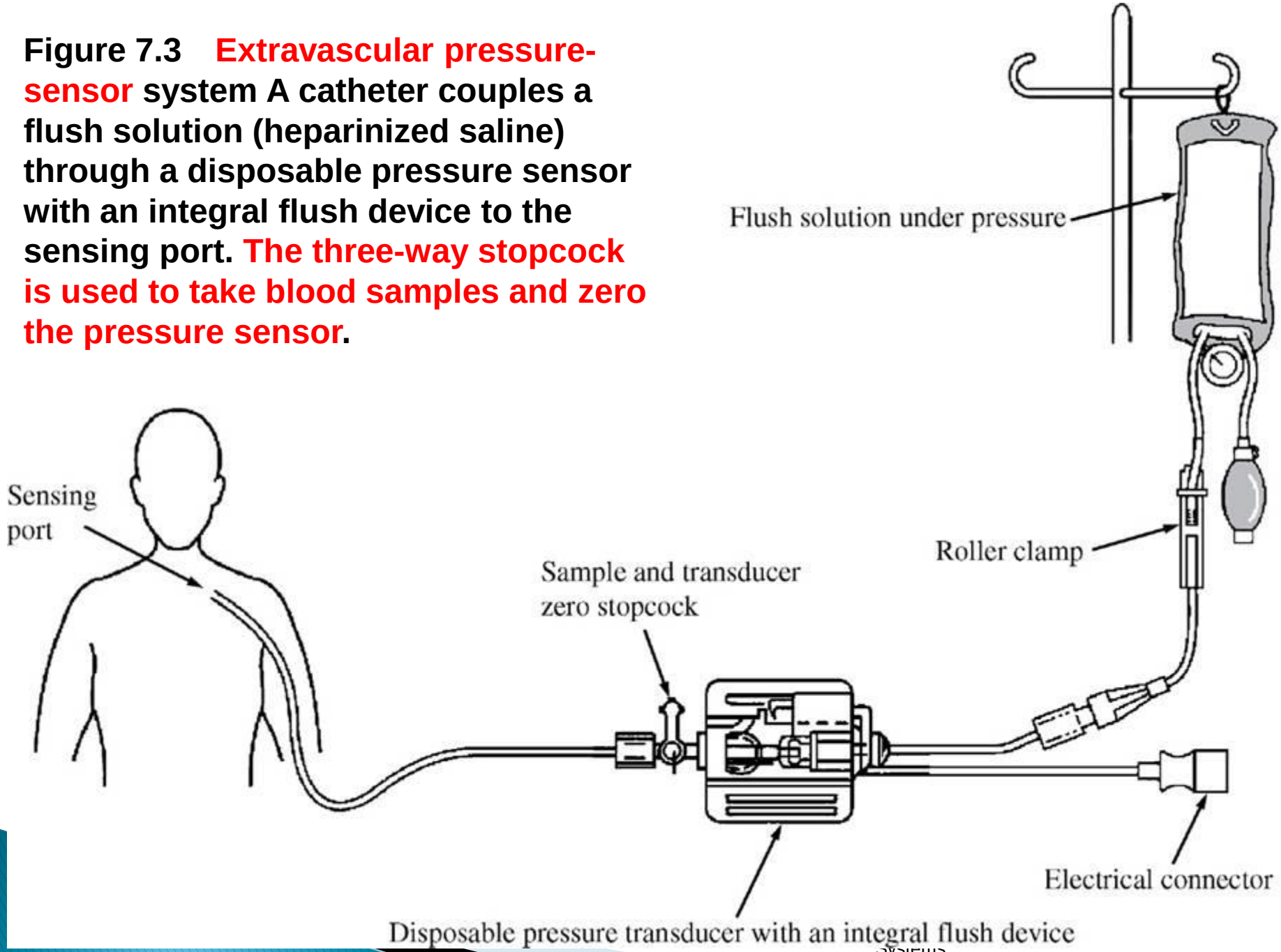
  國民健康局成人及中老年保健組科長陳姿伶表示，目前高血壓防治對象分成兩種族群，一種是血壓超過140 / 90mmHg，屬於需要藥物治療及生活型態改善之「高血壓患者」，這個族群會到疾病醫療照護體系裡；另一種就是在危險邊緣，還沒達到高血壓標準，卻瀕臨高血壓的高危險群，如血壓在120～139 / 80～89mmHg的人，這時候是表示血壓已經進入黃燈警示階段，但只要趕緊改變生活習慣，例如透過健康飲食、戒菸、規律運動等等，就能避免或延緩進入紅燈階段的高血壓時期。

Refer to http://www.doh.gov.tw/CHT2006/DM/DM1_p01.aspx?class_no=98&now_fod_list_no=8680&level_no=3&doc_no=47958

Direct measurement

- Blood-pressure sensor systems can be divided into two categories according to the location of the sensor element.
- The most common clinical method for directly measuring pressure is to couple the vascular pressure to an external sensor element via a liquid-filled catheter.
- The second category , the liquid coupling is eliminated by incorporating the sensor into the tip of a catheter that is placed in the vascular system. This device is known as intravascular pressure sensor.

Figure 7.3 **Extravascular pressure-sensor system A catheter couples a flush solution (heparinized saline) through a disposable pressure sensor with an integral flush device to the sensing port. The three-way stopcock is used to take blood samples and zero the pressure sensor.**



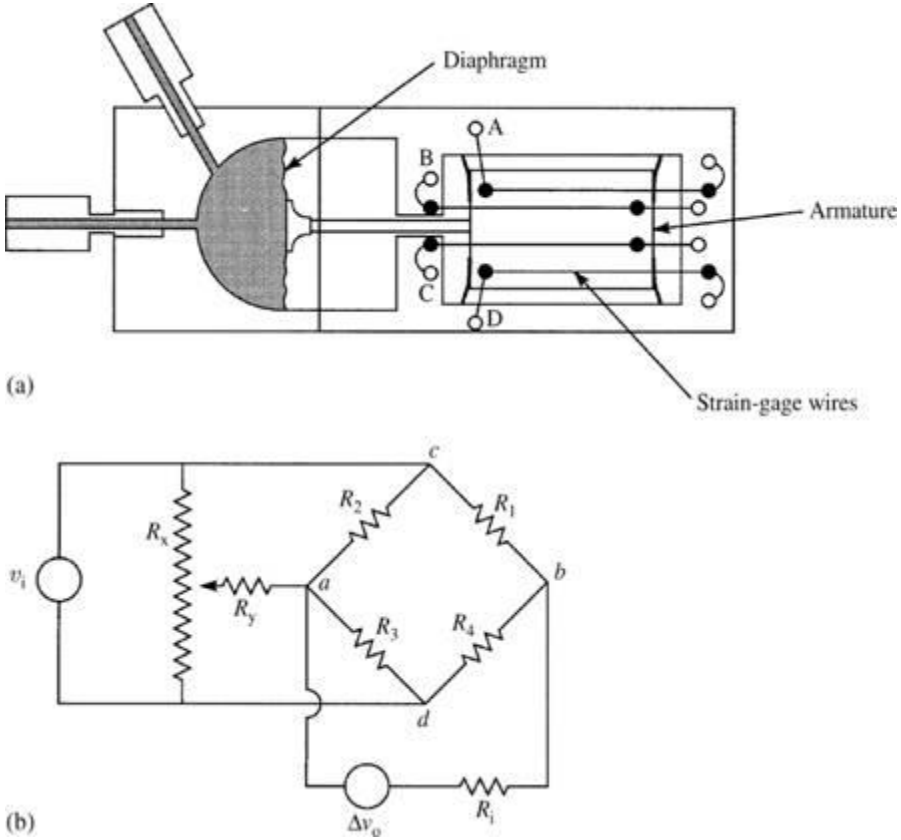


Figure 2.2 (a) Unbonded strain-gage pressure sensor. The diaphragm is directly coupled by an armature to an unbonded strain-gage system. With increasing pressure, the strain on gage pair B and C is increased, while that on gage pair A and D is decreased, (b) Wheatstone bridge with four active elements. *R*1 = B, *R*2 = A, *R*3 = D, and *R*4 = C when the unbonded strain gage is connected for translational motion. Resistor *Ry* and potentiometer *Rx* are used to initially balance the bridge, *vi* is the applied voltage and Δ*vo* is the output voltage on a voltmeter or similar device with an internal resistance of *Ri*.

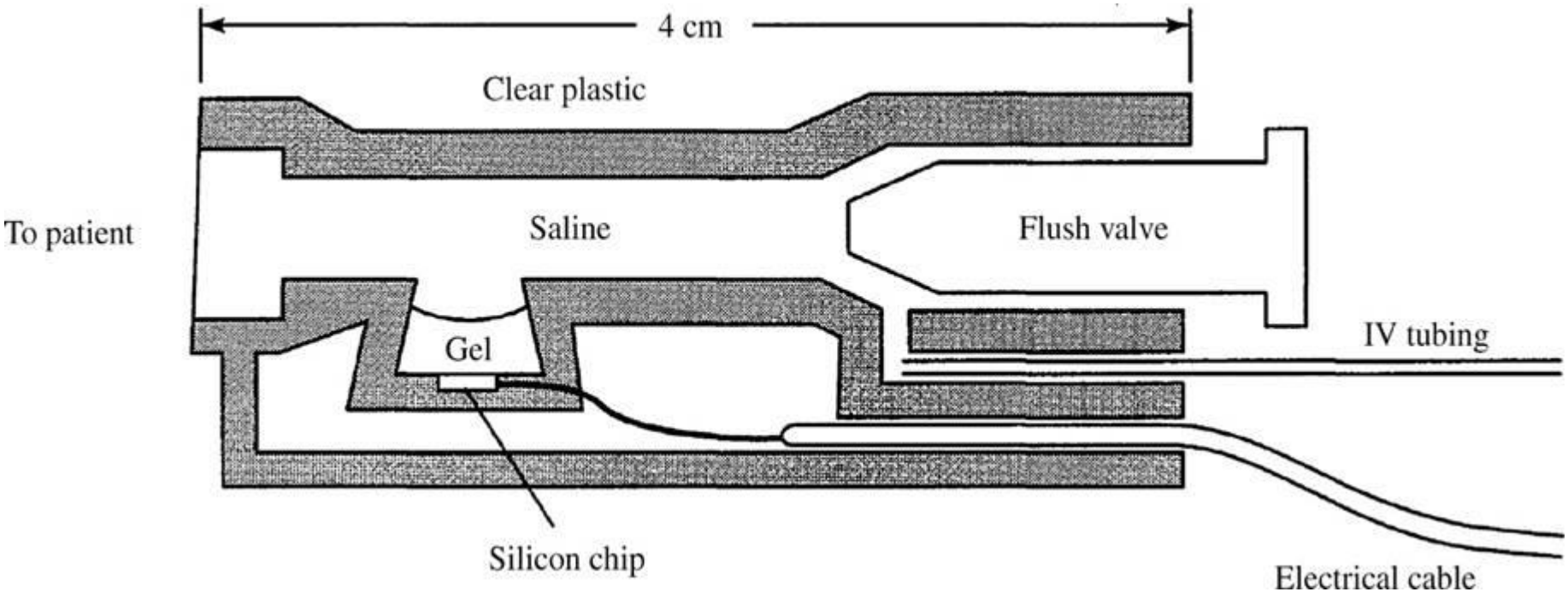
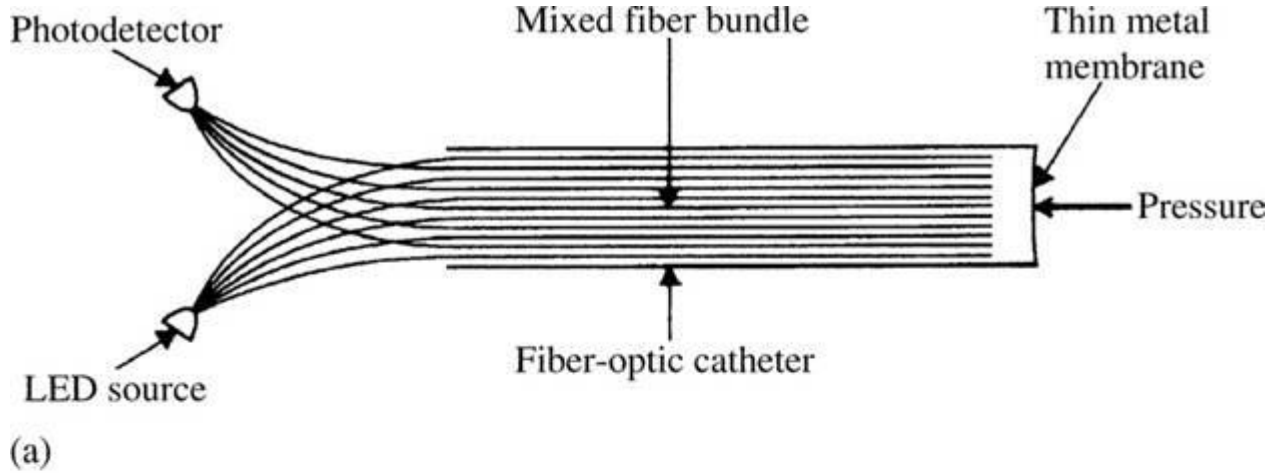


Figure 2.5 Isolation in a disposable blood-pressure sensor. Disposable blood pressure sensors are made of clear plastic so air bubbles are easily seen. Saline flows from an intravenous (IV) bag through the clear IV tubing and the sensor to the patient. This flushes blood out of the tip of the indwelling catheter to prevent clotting. A lever can open or close the flush valve. The silicon chip has a silicon diaphragm with a four-resistor Wheatstone bridge diffused into it. Its electrical connections are protected from the saline by a compliant silicone elastomer gel, which also provides electrical isolation. This prevents electric shock from the sensor to the patient and prevents destructive currents during defibrillation from the patient to the silicon chip.

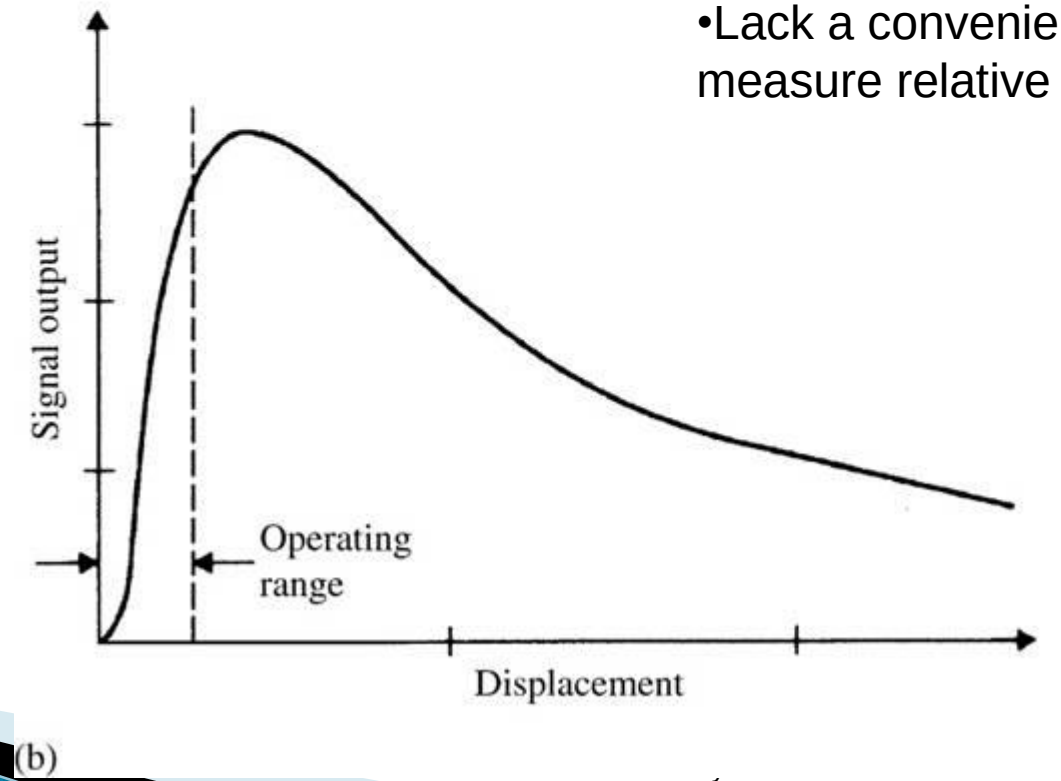
Intravascular sensors

- Detection of pressures at the tip of the catheter without the use of a liquid-coupling system can thus enable the physician to obtain a high frequency response and eliminate the time delay encountered when the pressure pulse is transmitted in a catheter-sensor system.
- A disadvantage of the catheter-tip pressure sensor is that it is more expensive than others and may break after only a few uses, further increasing its cost per use.

Figure 7.4 (a) Schematic diagram of an intravascular fiber-optic pressure sensor. Pressure causes deflection in a thin metal membrane that modulates the coupling between the source and detector fibers, (b) Characteristic curve for the fiber-optic pressure sensor.



- Safer electrically
- Lack a convenient way to measure relative pressure



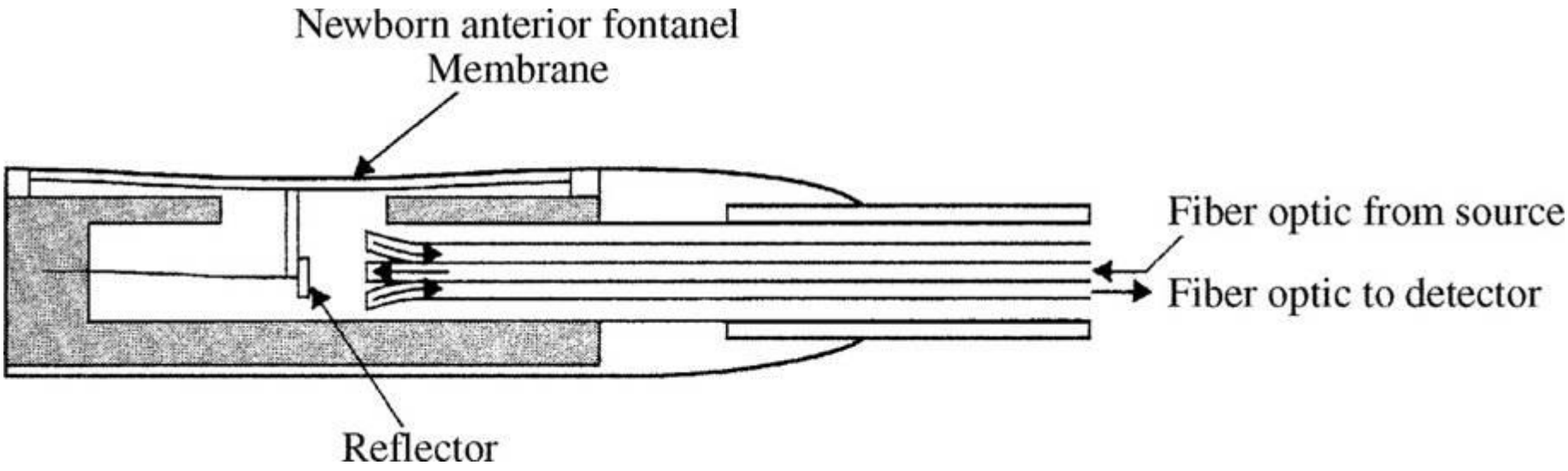
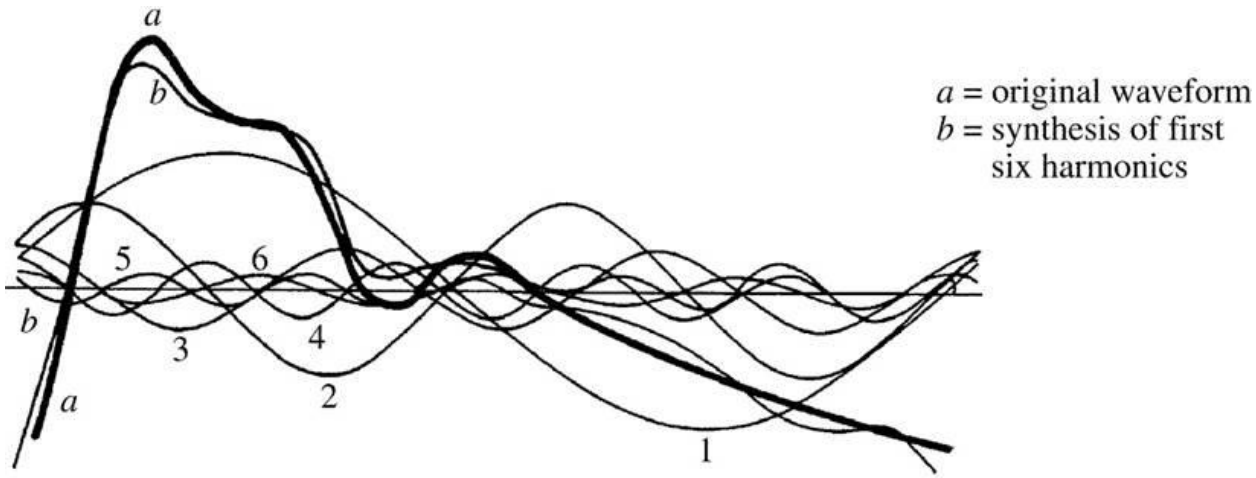


Figure 7.5 Fiber-optic pressure sensor for intracranial pressure measurements in the newborn. The sensor membrane is placed in contact with the anterior fontanel of the newborn.

Harmonic analysis of blood-pressure waveforms



Harmonic	Amplitude (%)
1	100
2	63.2
3	29.6
4	22.2
5	14.8
6	11.8

Figure 7.6 The first six harmonics of the blood-pressure waveform. The table gives relative values for amplitudes. (From T. A. Hansen, "Pressure Measurement in the Human Organism," *Acta Physiologica Scandinavica*, 1949, 19, Suppl. 68, 1–227. Used with permission.)

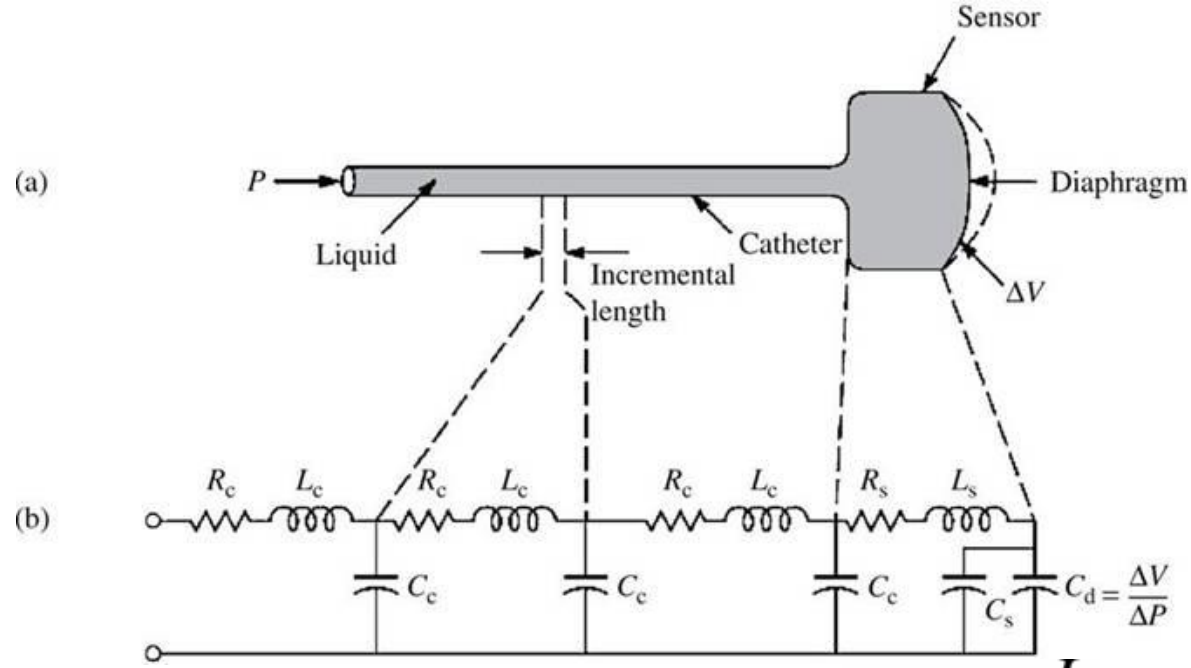
Dynamic properties of pressure-measurement systems

$$R_c = \frac{\Delta P}{F}\left(\mathrm{Pa \cdot s/m^3}\right)$$

$$R_c = \frac{\Delta P}{\bar{u}A}$$

$$R_c = \frac{8\eta L}{\pi r^4}$$

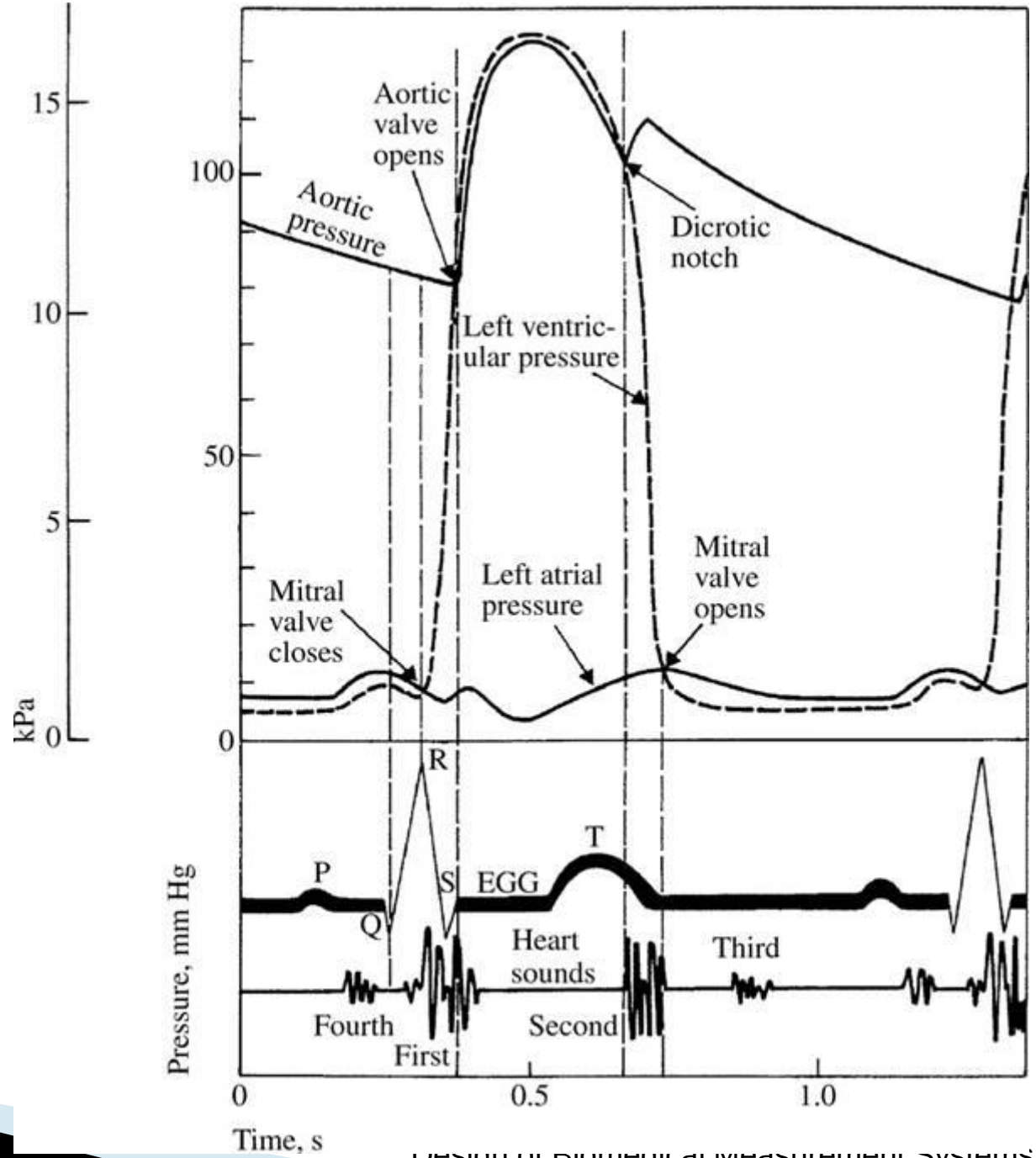
$$L_c = \frac{\Delta P}{dF/dt}\left(\mathrm{Pa \cdot s^2/m^3}\right)$$



$$L_c = \frac{\Delta P}{aA} \qquad L_c = \frac{m}{A^2} \qquad L_c = \frac{\rho L}{\pi r^2}$$

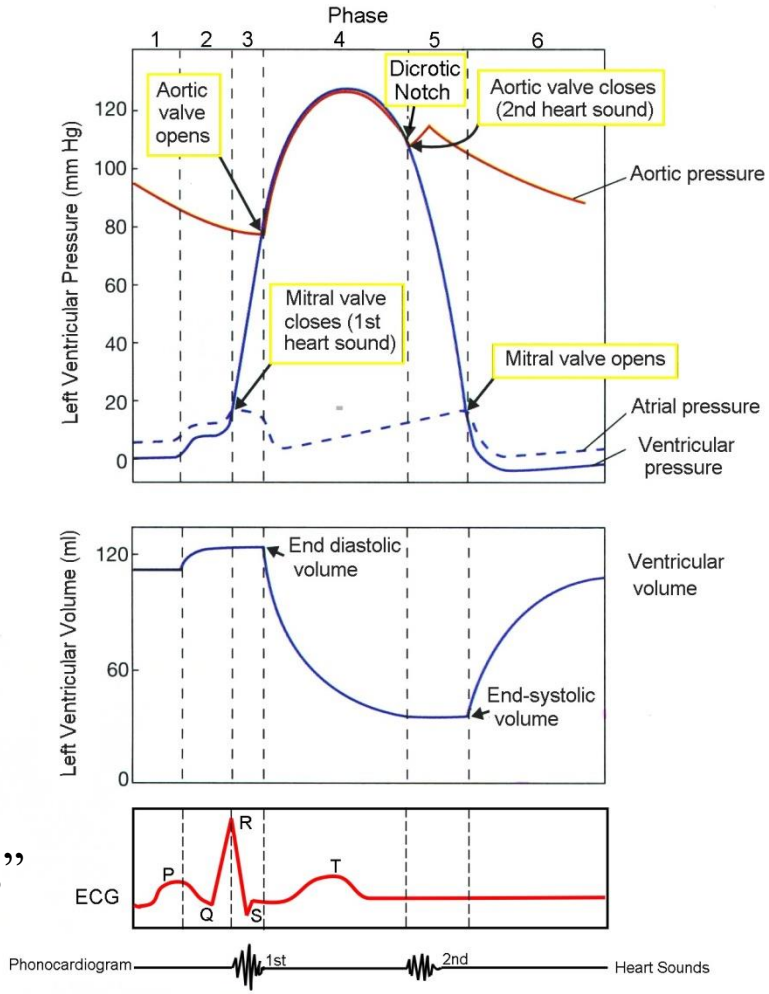
Figure 7.7 (a) Physical model of a catheter-sensor system, (b) Analogous electric system for this catheter-sensor system. Each segment of the catheter has its own resistance *R*c, inertance *L*c, and compliance *C*c. In addition, the sensor has resistance *R*s, inertance, *L*s, and compliance *C*s. The compliance of the diaphragm is *C*d.

Figure 7.15 Correlation of the four heart sounds with electric and mechanical events of the cardiac cycle.



Events of the Cardiac Cycle

- Events of the cardiac cycle.
  - Phase 1: Ventricular filling
  - Phase 2: Atrial contraction
  - Phase 3: Isovolumetric ventricular contraction
  - Phase 4: Ventricular ejection
  - Phase 5: Isovolumetric Ventricular relaxation
  - Phase 6: Ventricular filling
- Systole: phase 3 and phase 4, Diastole: phase 5, phase 6, phase1, and phase2.



Refer to R. Rhoades and R. Pflanzer, "Human Physiology," 3rd Edition, Saunders College Publishing, 1996.

Heart sounds

- The auscultation (聽診) of the heart gives the clinician valuable information about the functional integrity of the heart.
- Phonocardiography (心音描記術)
  - Temporal relationships between the heart sounds and the mechanical and electric events of the cardiac cycle.
- Heart sounds(心音)
  - vibrations or sounds due to the acceleration or deceleration of blood.
- Murmurs(心雜音)
  - vibrations or sounds due to blood turbulence

Heart sounds

- First heart sound
  - As the ventricles contract, blood shifts toward the atria, closing the atrioventricular valves with a consequential oscillation of blood.
  - Splitting of the first heart sound is defined as an asynchronous closure of the tricuspid and mitral values.
- Second heart sound
  - A low frequency vibration associated with the deceleration and reversal of flow in the aorta and pulmonary artery and with the closure of the semilunar values.

Heart sounds

- Third heart sound
  - Is attributed to the sudden termination of the rapid-filling phase of the ventricles from the atria and the associated vibration of the ventricular muscle wall, which are relaxed.
  - This low-amplitude, low-frequency vibration is audible in children and in some adults.
- Fourth or atrial heart sound
  - Is not audible but can be recorded by the phonocardiogram-occurs when the atria contract and propel blood into the ventricles.

Murmurs

- The source of most murmurs, developed by turbulence in rapidly moving blood, are know.
- Murmurs during the early systolic phase are common in children, and they are normally heard in nearly all adults after exercise.
- Abnormal murmurs may be caused by stenoses (狹窄) and insufficiencies (leaks)(閉鎖不全) at the aortic, pulmonary, and mitral values.

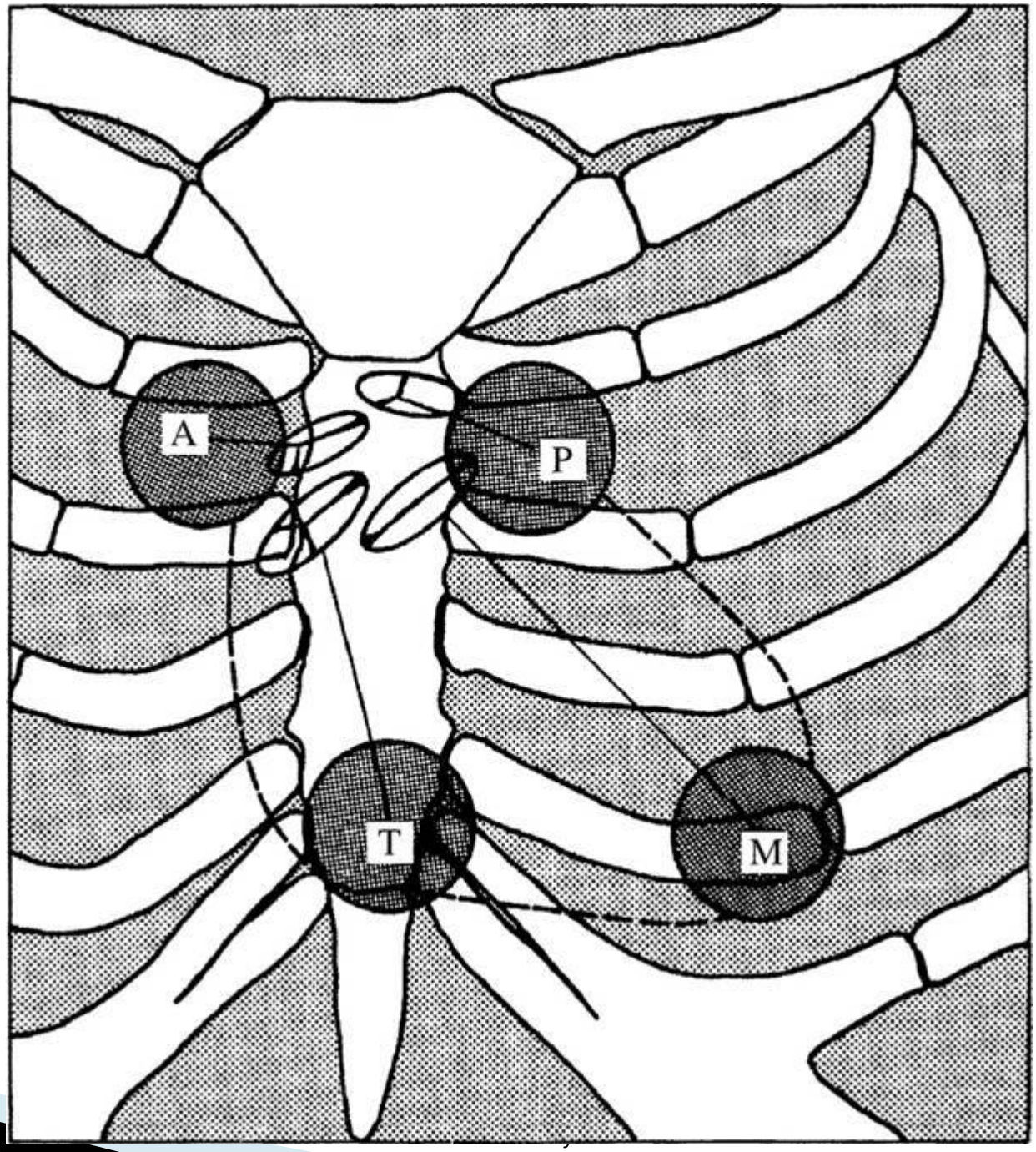
Auscultation techniques

- Heart sounds travel through the body from the heart and major blood vessels to the body surface.
- Because of the acoustical properties of the transmission path, sound waves are attenuated and not reflected.
- The largest attenuation of the wavelike motions occurs in the most compressible tissues, such as the lungs and fat layers.

Auscultation techniques

- There are optimal recording sites for the various heart sounds, sites at which the intensity of sound is the highest because the sound is being transmitted through solid tissues or through a minimal thickness of inflated lung.
- There are four basic chest locations at which the intensity of sound from the four valves is maximized. (Figure 7.16)

Figure 7.16 Auscultatory areas on the chest A, aortic; P, pulmonary; T, tricuspid; and M, mitral areas. (From A. C. Burton, Physiology and Biophysics of the Circulation, 2nd ed. Copyright © 1972 by Year Book Medical Publishers, Inc., Chicago. Used by permission.)



Auscultation techniques

- Heart sounds and murmurs have extremely small amplitudes, with frequencies from 0.1 to 2000 Hz.
  - Below about 20Hz, the amplitude of heart sounds is below the threshold of audibility.
  - Extraneous noise must be minimized in the vicinity of the patient.

Stethoscopes

- Stethoscopes are used to transmit heart sounds from the chest wall to the human ear.
- Some variability in interpretation of the sounds stems from the user's auditory acuity and training.
- Younger individuals revealed slighted better responses to a stethoscope than their elders.
- The critical area of the performance of a stethoscope (the clinically significant sounds near the listener's threshold of hearing) may be totally lost if the stethoscope attenuates them as little 3 dB.

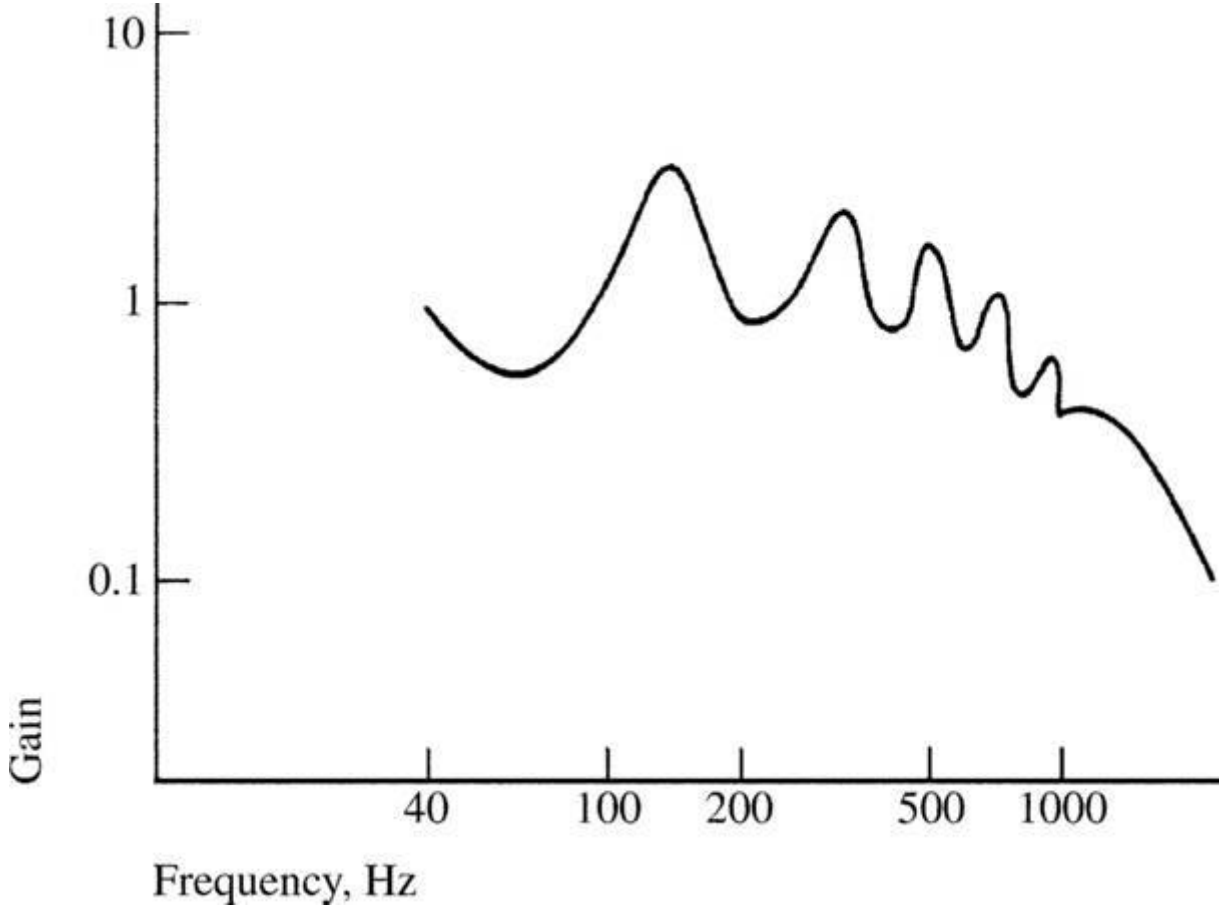


Figure 7.17 The typical frequency-response curve for a stethoscope can be found by applying a known audiofrequency signal to the bell of a stethoscope by means of a headphone-coupler arrangement. The audio output of the stethoscope earpiece was monitored by means of a coupler microphone system. (From P. Y. Ertel, M. Lawrence, R. K. Brown, and A. M. Stern, Stethoscope Acoustics I, "The Doctor and his Stethoscope." Circulation 34, 1966; by permission of American Heart Association.)

Stethoscopes

3M™ Littmann® Classic II S.E. Stethoscopes

3M™ Littmann® Electronic Stethoscope Model 4100WS with Ambient Noise Reduction

Electronic stethoscopes

- These devices have selectable frequency-response characteristics ranging from the "ideal" flat-response case and selected bandpasses to typical mechanical stethoscope responses.
- Physician, however, have not generally accepted these electronic stethoscopes,
  - mainly because they are unfamiliar with the sounds heard with them.
  - Their size, portability, convenience, and resemblance to the mechanical stethoscope are other important considerations.

Phonocardiography

- A phonocardiogram is a recording of the heart sounds and murmurs.
  - It eliminates the subjective interpretation of these sounds and also make possible an evaluation of the heart sounds and murmurs with respect to the electric and mechanical events in the cardiac cycle.
  - In the clinical evaluation of a patient, a number of other heart-related variables may be recorded simultaneously with the phonocardiogram.
    - These include ECG, carotid arterial pulse, jugular venous pulse, and apexcardiogram.

Indirect measurements of blood pressure

- 聽診測量技術(auscultation technique)
- 振盪測量技術(oscillometry)
- 壓張測量技術(tonometry)

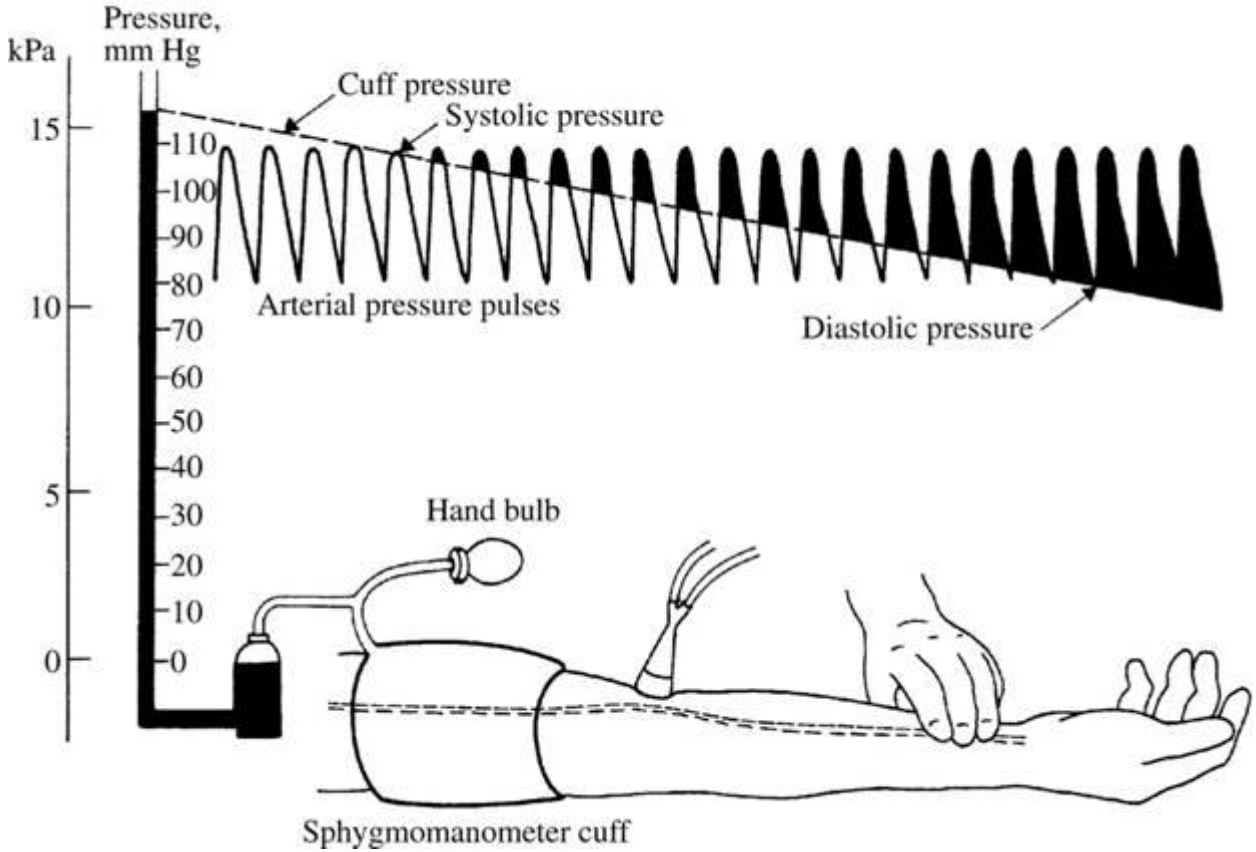


Figure 7.20 Typical indirect blood-pressure measurement system The sphygmomanometer cuff is inflated by a hand bulb to pressure above the systolic level. Pressure is then slowly released, and blood flow under the cuff is monitored by a microphone or stethoscope placed over a downstream artery. The first Korotkoff sound detected indicated systolic pressure, whereas the transition from muffling to silence brackets diastolic pressure. (From R.F. Rushmer, *Cardiovascular Dynamics*, 3rd ed., 1970. Philadelphia: W.B. Saunders Co. Used with permission.)

Auscultation technique

- Cuff size
  - 0.4 times the circumference of the extremity
- Cuff bled off speed
  - 2 to 3 mmHg/s
- Can not be used in a noisy environment, whereas the palpation technique can.
- The hearing acuity of the user must be good for low frequencies from 20 to 300 Hz.
  - The failure of the auscultation technique for hypotensive (低血壓) patients may due to low sensitively of the human ear to these low-frequency vibrations.

Automated auscultatory device

- Using a microphone replaces the stethoscope.
- Inflation with a rapid 20 to 30 mmHg/s to a preset pressure about 30mmHg higher than the suspected systolic level.
- The first Korotkoff sound is detected by the microphone, at which time the level of cuff pressure is stored.
- The muffling and silent period of the Korotkoff sounds is detected, and the value of the diastolic pressure is also stored.

Automated auscultatory device

- The ultrasonic determination of blood pressures employs a transcutaneous Doppler sensor that detects the motion of the blood-vessel walls in various states of occlusion.
- Advantages
  - Can be used with infants and hypotensive individuals and in high-noise environments.
- Disadvantages
  - Movements of the subject's body cause changes in the ultrasonic path between the sensor and the blood vessel.

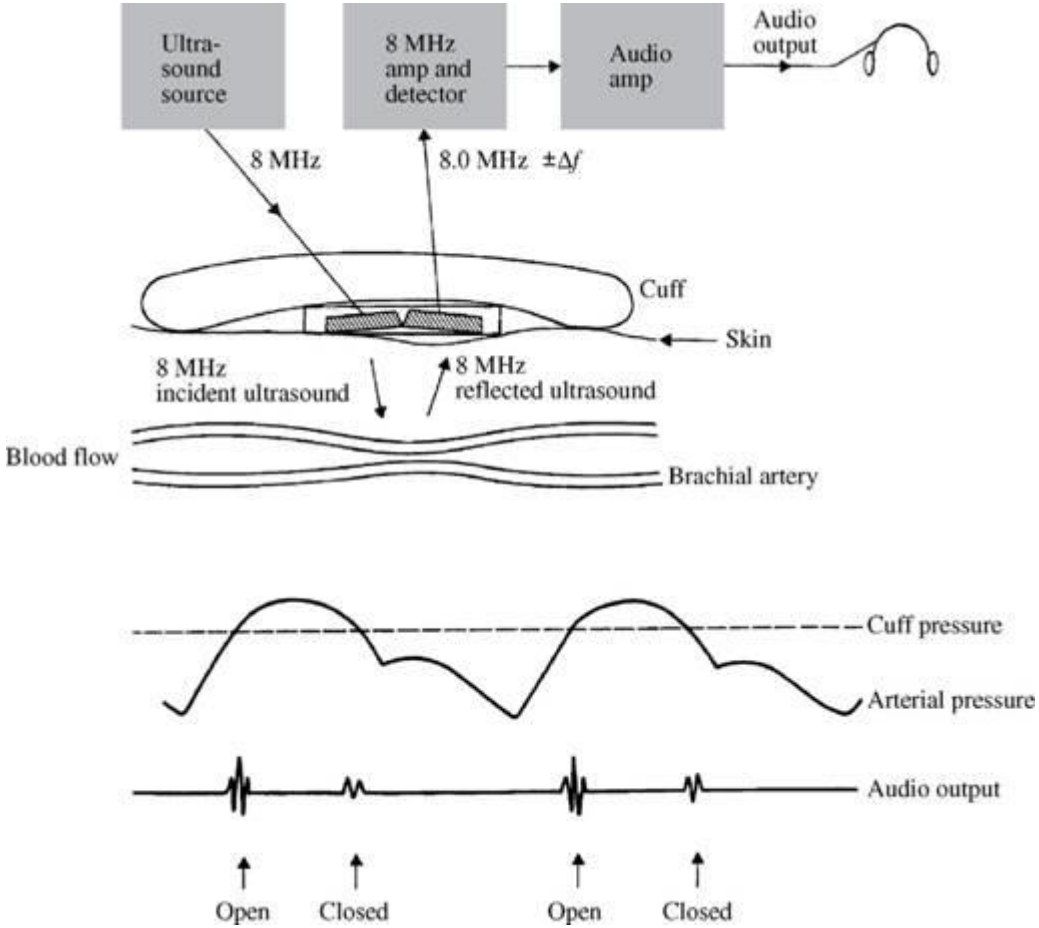


Figure 7.21 Ultrasonic determination of blood pressure A compression cuff is placed over the transmitting (8 MHz) and receiving (8 MHz ± Δ*f*) crystals. The opening and closing of the blood vessel are detected as the applied cuff pressure is varied. (From H. F. Stegall, M. B. Kardon, and W. T. Kemmerer, "Indirect Measurement of Arterial Blood Pressure by Doppler Ultrasonic Sphygmomanometry," *J. Appl. Physiol*., 1968, 25, 793-798. Used with permission.)

Oscillometric method

- The oscillometric method, a noninvasive blood pressure technique, measures the amplitude of oscillations that appear in the cuff pressure signal which are created by expansion of the arterial wall.
- The cuff-pressure signal increases in strength in the systolic pressure region, reaching a maximum when the cuff pressure is equal to mean arterial pressure.

Oscillometric method

- There is no clear transition in cuff-pressure oscillations to identify diastolic pressure since arterial wall expansion continuous to happen below diastolic pressure while blood is forced through the artery.
- Thus, oscillometric monitors employ proprietary algorithms to estimate the diastolic pressure.

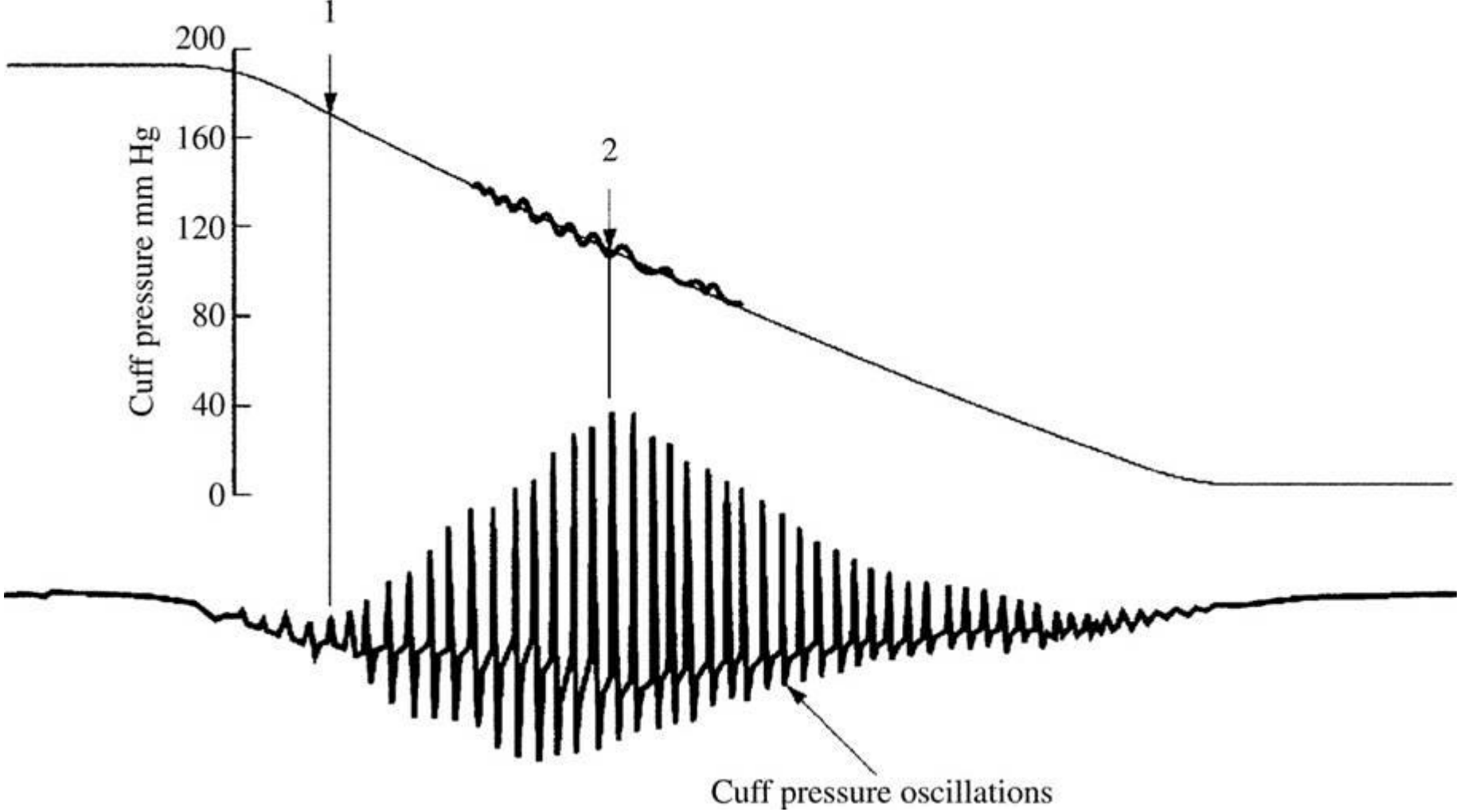


Figure 7.22 The oscillometric method A compression cuff is inflated above systolic pressure and slowly deflated. Systolic pressure is detected (Point 1) where there is a transition from small amplitude oscillations (above systolic pressure) to increasing cuff-pressure amplitude. The cuff-pressure oscillations increase to a maximum (Point 2) at the mean arterial pressure.

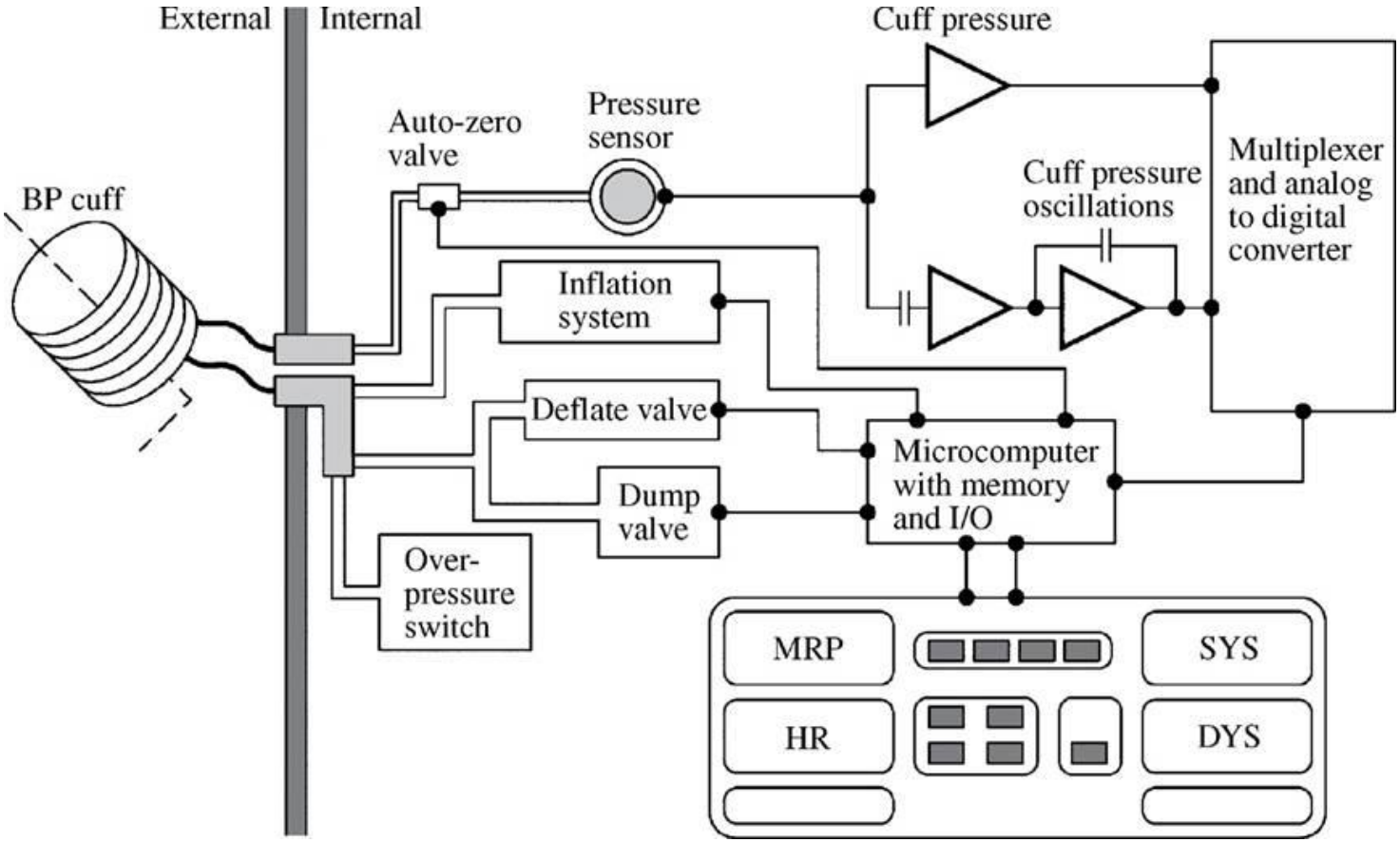


Figure 7.23 Block diagram of the major components and subsystems of an oscillometric blood-pressure monitoring device, based on the Dinamap unit, I/O = input/output; MAP = mean arterial pressure; HR = heart rate; SYS = systolic pressure; DYS = diastolic pressure. From Ramsey M III. Blood pressure monitoring: automated oscillometric devices, *J. Clin. Monit*. 1991, 7, 56–67.

Non-Invasive Blood Pressure Simulator

Fluke BP Pump 2

Fluke CuffLink

血壓計

- 水銀血壓計採用的是科氏音法，電子血壓計採用的是示波法，科氏音法是通過醫生用聽診器來聽血管內血液流動的聲音，示波法是通過測量血液流動時對血管壁產生的振動；雖然測量原理不同，但測試結果從理論上講是一樣的。不過，電子血壓計易受環境變數影響測量準確性，相對之下，水銀血壓計只要由訓練有素的醫護人員操作，所測出的結果都很準確。

Tonometry

- One significant advantage of the arterial tonometer is its ability to make noninvasive, non-painful, continuous measurements for long periods of time.
- The basic principle of tonometry is that, when a pressurized vessel is partly collapsed by an external object, the circumferential stresses in the vessel wall are removed and internal and the external pressure are equal.

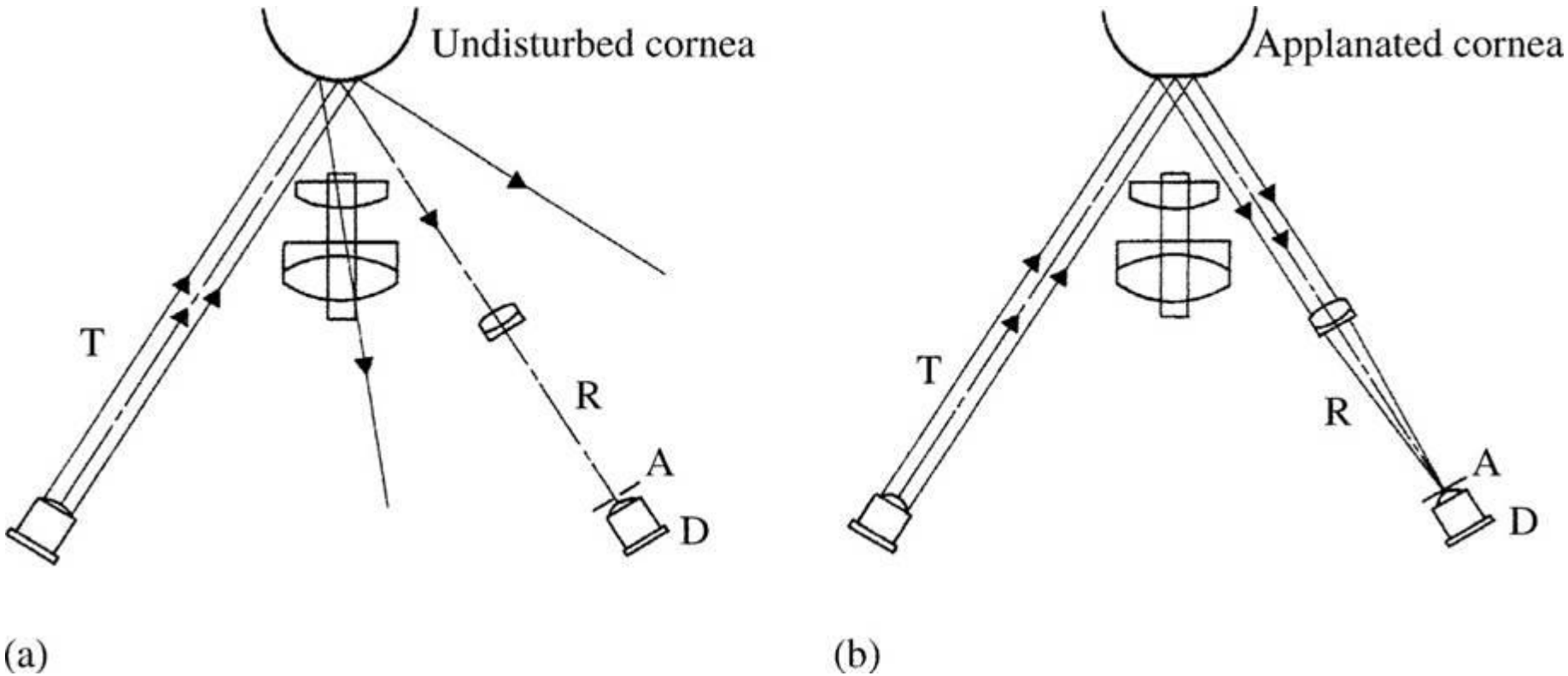


Figure 7.24 Monitoring system for noncontact applanation tonometer (From M. Forbes, G. Pico, Jr., and B. Grolman, "A Noncontact Applanation Tonometer, Description and Clinical Evaluation," *J. Arch. Ophthalmology*, 1975, 91, 134–140. Copyright © 1975, American Medical Association. Used with permission.)

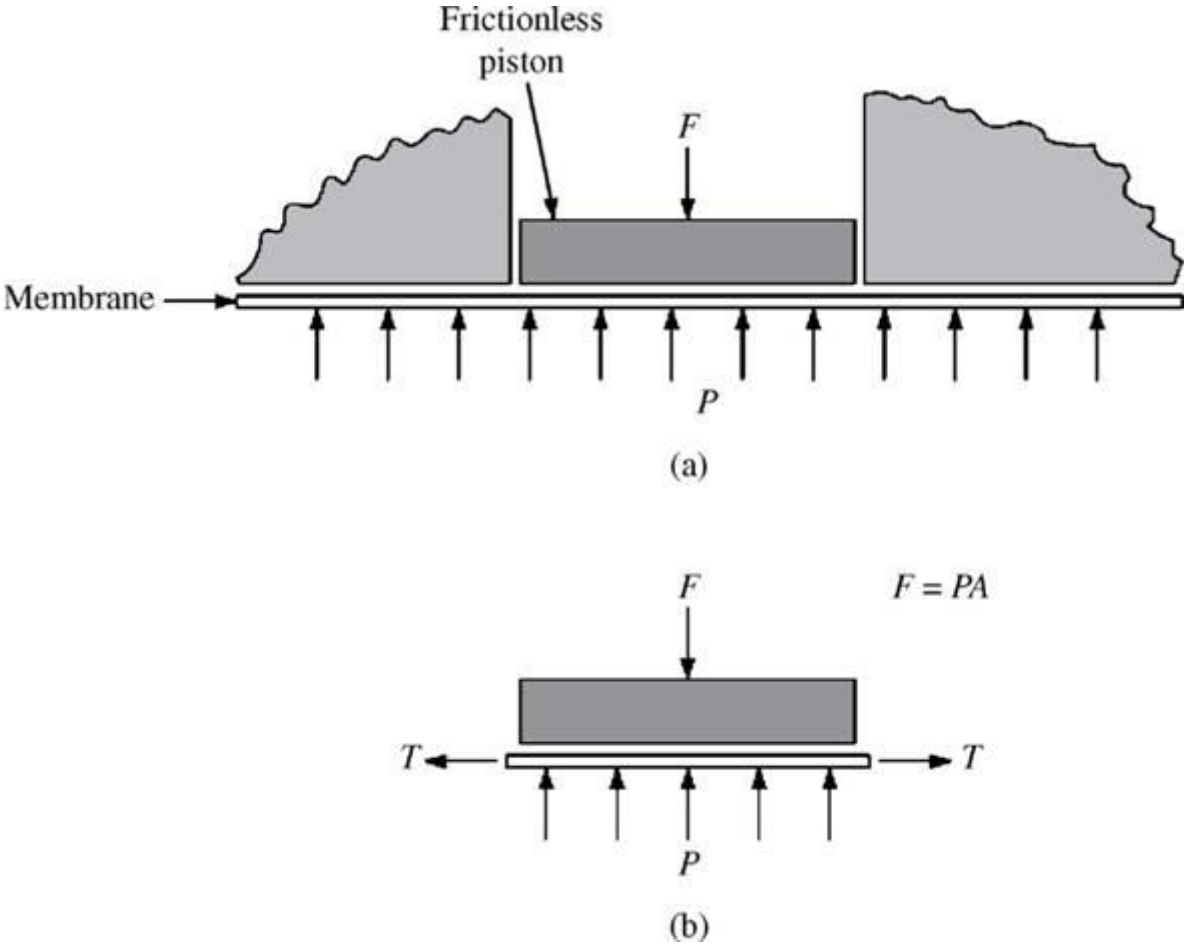


Figure 7.25 Idealized model for an arterial tonometer, (a) a flattened portion of an arterial wall (membrane). *P* is the blood pressure in a superficial (表面的)artery, and *F* is the force measured by a tonometer transducer, (b) a free-body diagram for the idealized model of (a) in which *T* is the membrane tensile force perpendicular to both *F* and *P*. From Eckerle, J. D., "Tonometry, arterial," in J. G. Webster (ed.), *Encyclopedia of Medical Devices and Instrumentation*. 2nd ed. New York: Wiley, 2006, vol. 6, pp. 402–410.

Tonometry

- Using a single arterial tonometer, a major practical problem is that the arterial rider must be precisely located over the superficial artery.

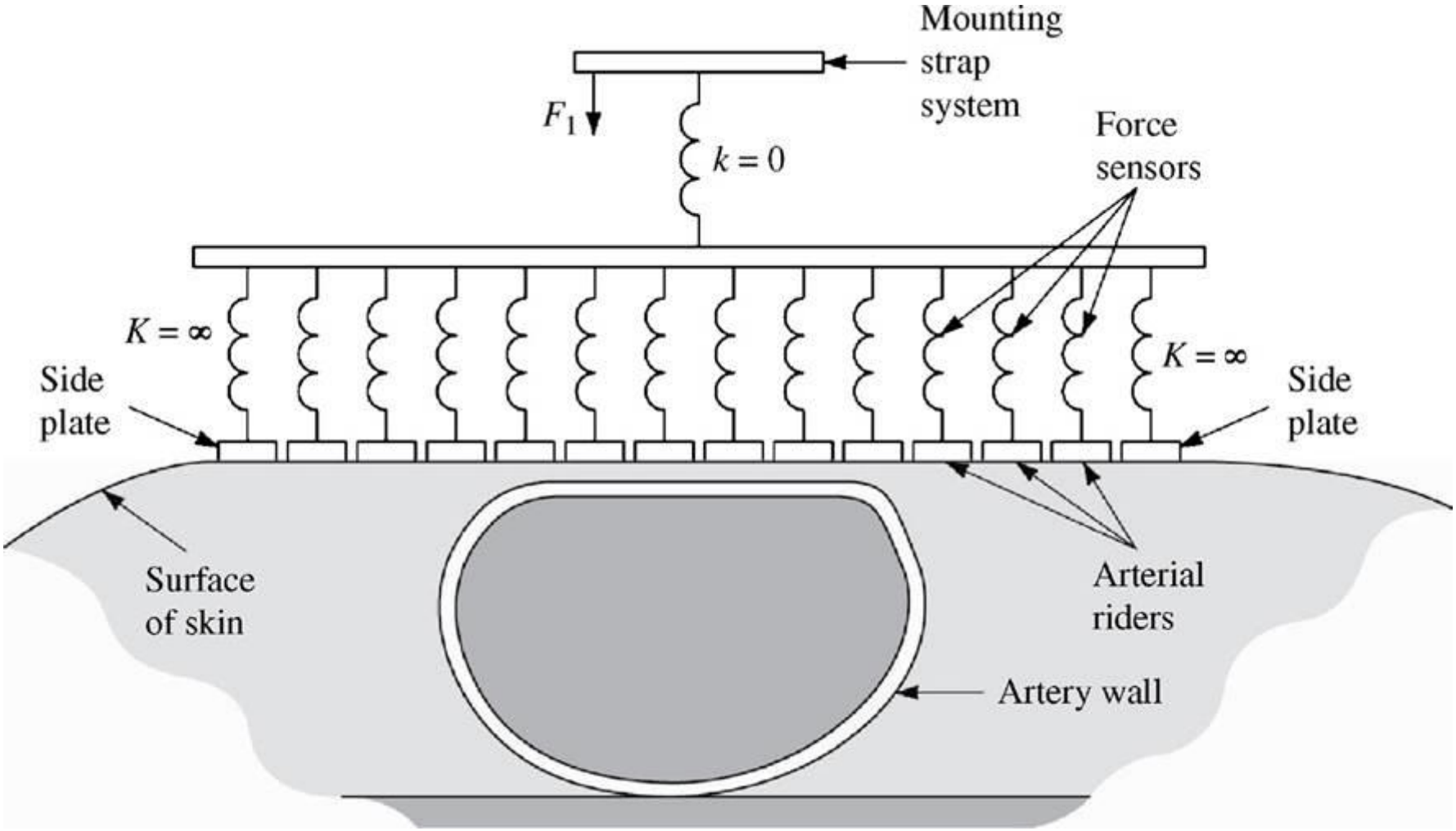


Figure 7.26 **Multiple-element arterial tonometer. The multiple element linear array of force sensors and arterial riders are used to position the system such that some element of the array is centered over the artery. From Eckerle, J. D., "Tonometry, arterial," in J. G. Webster (ed.), *Encyclopedia of Medical Devices and Instrumentation*. 2nd ed. New York: Wiley, 2006, vol. 6, pp. 402–410.**